ML01 Your first instruction and your first prediction

Week 1, Python and a first prediction | Session 1 of 24 | Machine Learning | New library: pandas, sklearn

Today's question. Can a computer guess a rent after seeing 120 homes?

Objective. By the end: you can run, save and reopen a notebook, and you can say what a prediction is using the rent example.

> **WHERE THIS IS USED**
> Property apps and agents guess a fair rent from the size and area of a home. A model does the same after studying homes that were already rented, and today you build the smallest version. What rent would you guess for a home you know?

Before this session: START_HERE.pdf. No subject knowledge is assumed. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
file	A named container for your work on the laptop.	A paper file in a folder.
notebook	A page made of text cells that explain and code cells that run.	An exercise book where the answers can run themselves.
code cell	A box where you type Python and press Shift+Enter to run it.	A calculator that sits inside your page.
output	What Python shows under a cell after it runs.	The answer a calculator shows after you press equals.
print	A ready-made instruction that shows a value on the screen.	Reading a number out loud.
prediction	A number or label a computer guesses for something it has not seen yet.	A fundi estimating a job before measuring it.

Block A. Run your first instruction

Open ML01.ipynb in Jupyter. Click the first code cell and press Shift+Enter.

Predict. Write on paper what you think these two lines will show.

Your guess, written before you run it

```
print("Hello, data")
print(3 + 2)
```

Run it, then compare. The first line shows Hello, data and the second shows 5. A notebook is a page of text cells and code cells. Shift+Enter runs a code cell and its output appears underneath. print is a ready-made instruction, and the brackets hold what it shows.

Investigate.

1. What happens if you click a text cell and press Shift+Enter?
2. Why does Hello, data sit inside quotation marks while 3 + 2 does not?

Modify. Change Hello, data to your own nickname and run the cell again.

Block B. Run a ready-made prediction

This model has looked at 120 invented homes and learned how rent follows floor area. You do not need to understand it yet. Session 13 opens it up.

Predict. Before you run it, write the monthly rent in UGX you expect for a home of 50 square metres. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
from sklearn.linear_model import LinearRegression
homes = pd.read_csv("synthetic_homes.csv")
model = LinearRegression().fit(homes[["area_m2"]], homes["rent_ugx"])
pred = model.predict(pd.DataFrame({"area_m2": [50]}))[0]
print("Estimated rent for 50 square metres:", round(pred))
```

Run it, then compare. The model estimates about UGX 309,260 a month. It drew a line through the 120 homes and read the rent at 50 square metres. Guessing a number from examples is called regression, and you will build one yourself in session 14. The homes are invented, so the estimate says nothing about real rents.

Investigate.

1. The home grows from 50 to 100 square metres. What do you expect the estimate to do?
2. Is the estimate certain?
3. Does this model know anything about Kampala?

Modify. Change 50 to 100 and run the cell again.

Make

In a new code cell print your nickname and the result of 7 + 4. In another cell estimate the rent for 80 square metres. Save as ML01.ipynb, close it and reopen it. On your progress map, write one topic you love, such as football, music or farming. It will grow into your project. With the instructor, read AI_USE_POLICY.pdf: in weeks 1 to 6 all core work is your own, with no AI-generated solutions.

Save your work as ML01. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
SyntaxError or invalid character	A quotation mark is curly, usually from copying out of a document.	Retype the quote marks with the keyboard so they are straight.
ModuleNotFoundError: No module named 'sklearn'	This laptop is not using the prepared camp environment.	Tell the instructor. Do not install anything yourself.

Exit ticket

A prediction is a guess for something not yet seen; the rent model guesses a number from 120 examples.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.